Building Rooftop Extraction from High-Resolution Aerial Images

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Abstract

Rooftop extraction of buildings plays a significant role in Urban Planning, Smart City as well as Cadastral Mapping and Environmental Monitoring. The availability of very high-resolution aerial imagery has substantially facilitated building detection and analysis accompanied however by some challenges such as scale variation, occlusions, and spectral similarities between targets and their background particularly between roofs and roads made from the same material (Huang et al., 2020). This report discusses rooftop extraction from aerial imagery with a particular focus on Athens motivated by the recent increasing demand for accurate building footprint detection in the context of sustainable urban development which requires accurate rooftops to build reliable geospatial databases. The case study area was chosen to be Athens because it features relatively homogeneous as well as flat rooftop structures thus easier extraction process plus cadastral footprints are available to be used as ground truth. This project does not implement the use of MSGP and SCRD though it recognizes their potential contributions toward reaching higher levels of accuracy as discussed. The results prove practicability in extracting rooftops from aerial images of Athens which can later on be integrated with smart urban systems.

1. Introduction

The extraction of roofs from buildings has its major applications in urban planning, cadastral mapping, and smart city planning with environmental monitoring. Due to the growing availability of very high-resolution aerial images, building footprint detection becomes easier and attainable at a greater scale, providing strong support for reliable geospatial databases. Accurate geospatial data is encouraged so that infrastructure monitoring, solar energy assessment, or disaster response planning applications can be developed. Remote sensing automation methods now perform much faster and more consistently than any traditional manual approach.

Even with these technological improvements, issues such as scale variation, shadows, vegetation occlusion, and spectral similarity between rooftops and other surfaces, such as roads, still account for some of the major barriers.(Xu et al., 2019, Bittner et al., 2019) Recently, researchers have ventured into the implementation of the Multi-Scale Global Perceptron (MSGP) and the Spatial Context Refinement Decoder (SCRD), which are advanced methods for solving boundary roughness and facilitating both global and local building features.(Yuan et al., 2025) This project does not make use of MSGP or SCRD.

The study area selected for this research is Athens (Figure 1), Greece. The already existing cadastral footprints can be a good source of ground truth data to check the results. Such features of Athens not only make the city a grounded setting to test out different methods of roof air extraction, but also allow the findings to be less limited to the local area but to be applicable to other more diverse and heterogeneous urban areas.

Figure 1. Boundary of Athens

2. Motivation

The need for precise rooftop identification has been notably increasing, mainly due to the need for high-quality geospatial data sets that are used in the urban management and planning sectors. Since cities are developing, the establishment of a new spatial base, as well as updating building footprints, are the primary areas where the demand for cadastral records and the corresponding infrastructure planning for sustainable development have arisen. Besides, such manual mapping techniques that are carried out by humans can be said to suffer from errors and they are labor-intensive and time-consuming. Automated extraction methods that incorporate high-resolution imagery can alleviate such drawbacks by offering quicker and more dependable solutions. Additionally, understanding where roofs are located can give valuable insights to the likes of energy modeling, clean energy assessments i.e. the installation of solar panels and risk evaluation in places that are susceptible to natural disasters. The present-day supply of urban data integration of remote sensing and artificial intelligence is one of the

key indications of the coming to be of the subject matter of this paper.

3. Literature Review

3.1 Traditional Approaches for Building Extraction

The early research documented powerfully the use of handcrafted features in building detection from aerial images. The properties of the spectra, the structure and the texture of the remote sensing data were the main characteristics used for feature extraction. These features along with threshold-based methods allowed limited success in identifying a building's rooftop from the rest of the land cover. Nevertheless, such methods were very sensitive to lighting changes, shadows, and vegetation which affected the method's performance in different urban atmospheres. Besides, they could not generalize across different building sizes and heterogeneous backgrounds, which indicated that more robust solutions were necessary.

3.2 CNN-Based Feature Learning Methods

Deep learning has made a major breakthrough in the use of Convolutional Neural Networks (CNN) for rooftop extraction tasks. The development of Fully Convolutional Networks (FCN) and their derivatives allowed the implementation of pixel-wise classification. For example, Bittner et al. (2018) suggested multi-branch CNNs that merge RGB, nDSM, and panchromatic inputs for better feature separation. Besides, Shi et al. (2020) applied graph-based CNNs for preservation of sharp building boundaries. However, as a result of these advancements, CNNs were still in a position to face several challenges with multi-scale variability and limited receptive fields, which often caused them to produce fragmented or incomplete building predictions.

3.3 Attention-Enhanced CNN Frameworks

To mitigate the limitations of CNN, attention mechanisms were proposed to optimize spatial and channel-wise information. Huang et al. (2019) proposed residual FCNs with the gate encoder to remove redundant features while some other researchers worked on adaptive edge modules and affinity-based learning for salient object detection. These methods made it better preserved details at the boundaries however, since they are based on local convolutional filters, long-range semantic dependencies were not adequately explored.

3.4 Transformer-Based Methods

One of the recent innovations in Vision Transformers (ViT) is the addition of global context modeling into rooftop extraction. Unlike CNNs, Transformer through the process of self-attention are able to keep track of relationships anywhere in the image. To illustrate, Wang et al. (2021) came up with window-based attention to have a good computational efficiency balance, while Chen et al. (2021) made use of multi-view CNN–Transformer fusion for better road and building segmentation. Swin Transformer and SegFormer became the main frameworks that were able to provide hierarchical multi-scale feature learning. The problem with these models was that they were quit often unaware of the fine-grained details at object boundaries, which consequently caused fragmentation and spectral confusion when these models were used for complex urban datasets.

3.5 Hybrid CNN–Transformer Architectures

Several hybrid models have been introduced that exploit the complementary strengths of CNNs (local detail extraction) and Transformers (global semantic representation) such as Swin-Unet that enabled local window attention along with skip connections for hierarchical fusion or Pyramid Vision Transformer (PVT) that facilitated global context through pyramid-based multi-scale attention. Though, the majority of these methods had depended on simple feature fusion strategies for the most part, which only allowed them to go a little way in refining shallow features. Consequently, the problems of capturing scale variations and decreasing the number of objects that can be mistaken for buildings such as roads and vegetation still remained.

3.6 Towards Multi-Scale Global Perceptron and Context Refinement

The limited CNN and Transformer based methods highlight the necessity of more efficient implementation of multi-scale context and feature refinement. One of such recent developments is the Multi-Scale Global Perceptron (MSGP) module that fills up this gap by combining depth wise separable convolutions for multi-scale local feature with a Transformer-based global attention. Besides, the Spatial Context Refinement Decoder (SCRD) not only supports the hierarchical fusion but also empowers it by re-calibrating the shallow features with the deep semantic tokens. These novel features are addressing the problems of fragmentation, spectral heterogeneity, and detail restoration that are the main causes of rooftop extraction, thus they become the most innovative solutions in this area.

4. Methodology

High quality data preparation forms the backbone of deep learning - based geospatial analysis. In the case of rooftop extraction from aerial imagery, the dataset must undergo careful refinement to ensure that models learn from accurate, representative and consistent samples. The following subsections outline the dataset acquisition, preprocessing, cleaning and encoding strategies adopted in this study.

4.1 Data Acquisition

The study area covering the urban extent of Athens, Greece was delineated using QGIS (version 3.16). High resolution base imagery was retrieved through the QuickMapServices plugin with Google Satellite as the source to provide detailed surface representation. All spatial processing was carried out in the WGS 84 coordinate system (EPSG: 4326). Building footprint data for the same region were obtained from Geofabrik (Open Street Map) and used to clip the city boundary to the required extent. To enable spatial sampling and model training, a 100 x 100 m grid with 512 x 512 pixel was generated with the QGIS Atlas tool (Figure 2).

This workflow produced two complementary data sets:

1. Raster tiles: Satellite imagery covering the gridded study area
2. Vector layers: Building footprint shapefiles used as training label

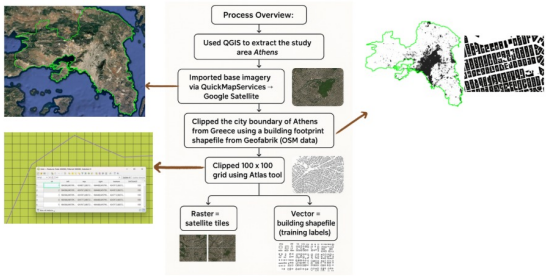

Figure 2. Rooftop extraction workflow in Athens

4.1.1 Challenges of Data:

Although a structured workflow was implemented to acquire and preprocess the Athens dataset, several characteristic limitations of urban geospatial data were identified. These issues, stemming from heterogeneous urban environment and the use of openly sourced datasets have implications for the accuracy and robustness of subsequent analyses.

1) Building footprint misalignment: A consistent horizontal offset of approximately 10 cm was detected between OpenStreetMap (OSM) building footprints and the Google Satellite base imagery. Such errors, arising from variations in digitization standards and coordinate transformations, were corrected by shifting the building footprints (Figure 3) to achieve proper alignment, ensuring spatial accuracy for subsequent analyses (Zhou and Cai, 2018).

Figure 3. Overlay of original footprint (yellow) and offset (red) building footprint (right) illustrating displacement from reference imagery (left)

2) Presence of non–building grids: Some 100×100 m grid cells primarily contained vegetation or open ground. While these tiles contained more than 500 labeled pixels, they contributed minimal building-related information, reflecting the natural heterogeneity of the urban landscape.

These identified limitations underscore the need for careful preprocessing to maintain data integrity in subsequent analyses.

4.1.2 Data Filtering:

To address the issue of irrelevant image tiles lacking the building information, a systematic preprocessing procedure was implemented (Figure 4). Building footprints, initially represented as polygons were converted into centroid points to enable precise localization of buildings within each grid. A counting operation was then performed to determine the number of buildings centroids per image tile. Based on this measure, a threshold was applied whereby grids containing fewer than five centroids were excluded from the dataset, while those exceeding this threshold were retained. This filtering process ensured that only image tiles with sufficient building density were incorporated into subsequent analyses (Sun and Zhou, 2022), thereby improving the computational efficiency for model training.

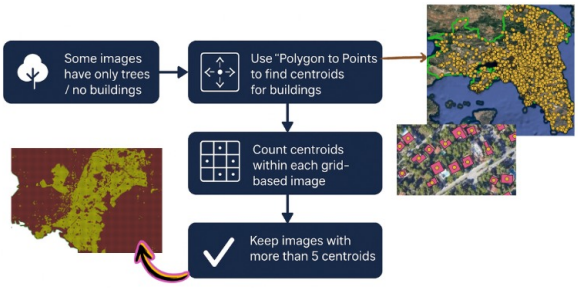

Figure 4. Data Filtering Workflow

4.1.3 Data Encoding and Model Ready Transformation

Following the spatial filtering and alignment procedures, the processed dataset was transformed into a format suitable for deep learning. Each tile consisted of a 512 x 512 pixel RGB image and a corresponding segmentation mask derived from the building footprints. To enable efficient training, the label masks were converted into a one hot encoded representation with two channels.(Figure 5)

1) $[1, 0]$ corresponding to background pixels

2) $[0, 1]$ corresponding to rooftop pixels

This encoding scheme ensured that the neural network could differentiate between classes on a per pixel basis while allowing the use of categorical loss functions for optimization.

Figure 5. Building segmentation – aerial image (left) and two class label (right)

During dataset inspection, a challenge was identified where some masks exhibited an unintended third class. This anomaly was attributed to digitization artifacts, shadow effects and boundary uncertainties at rooftop edges. While these cases did not dominate the dataset, they introduced label error that could affect convergence. To address this, all label three pixels were reclassified into to label two pixels, thereby restoring the intended binary classification scheme. This correction ensured that the dataset remained consistent across all samples and avoided unnecessary complexity in the training phase. The resulting dataset was thus both uniform in class structure and optimized for binary rooftop extraction (He and Yao, 2021)(He et al.,2021) forming a reliable foundation for subsequent model training and evaluation.

4.2 Data Processing

The data used in this study underwent several preparation steps to ensure quality, balance and compatibility with the deep learning framework. The following subsections describe the dataset preparation, model architecture and training setup in detail.

4.2.1 Dataset Preparation

After collection and cleaning, the dataset was prepared for deep learning through a structured workflow. Each 512x512 pixel RGB aerial image was paired with its corresponding rooftop mask and organized into separate directories for training, validation and testing. This organisation ensured reproducibility and prevented spatial leakage thereby keeping performance evaluation fair and realistic. To make the labels suitable for machine learning, rooftop masks were converted into binary one-hot encoded arrays where each pixel was assigned the value [1, 0] if it belonged to the background or [0, 1] if it corresponded to a rooftop. This encoding scheme was directly aligned with the network's softmax activation in the output layer and allowed optimization through binary cross-entropy loss ensuring that the model learned to distinguish between the two classes in a probabilistic manner. At the same time, tiles heavily dominated by background were removed since their addition would have made the dataset unbalanced. The resulting dataset consisting of approximately 19,900 samples provided a more balanced distribution of rooftops and non-rooftop areas allowing the network to focus on meaningful urban features instead of defaulting to background predictions

4.2.2 Model Architecture

The processed dataset was loaded into a TensorFlow/Keras pipeline within the Google Colab environment. This pipeline was responsible for efficient loading, shuffling and batching of data ensuring optimal utilization of computational resources during training. The model itself was implemented as a convolutional neural network (CNN) with an encoder–decoder design (Figure 6), a widely adopted structure in semantic segmentation (Ronneberger et al., 2015) tasks due to its capacity to learn both local and global representations of imagery. In this structure, the CNN encoder acted as the feature extractor reducing spatial resolution at each stage while progressively capturing more abstract and semantically meaningful information.

The process began with a convolutional layer using 32 filters and the ReLU activation function enabling the network to pick up simple visual details such as edges, corners and textures. This was followed by a max pooling step which reduced the feature maps from 512 × 512 to 256 × 256 making the data more compact while retaining the most useful information. A second convolutional block, this time with 64 filters was then applied again followed by pooling which further reduced the spatial resolution to 128 × 128 and enabled the extraction of more abstract rooftop descriptors. At the bottleneck, a deeper Conv2D layer with 128 filters was used to capture high-level semantic features(LeCun et al., 2015) equipping the network with the capacity to recognize rooftops even when they were partially occluded by vegetation shadowed by neighbouring structures or spectrally similar to surrounding paved surfaces.

Following the encoder, the decoder progressively reconstructed the segmentation mask back to its original spatial resolution. This was achieved using transposed convolutional layers (Conv2DTranspose) often referred to as deconvolutions which perform learned upsampling. The first of these equipped with 64 filters expanded the feature maps back to 256 × 256 followed by a second with 32 filters that restored the dimensions to 512 × 512. To complete the process, a 1 × 1 convolutional layer with softmax activation was employed to map each pixel's feature vector into two probability values, one for rooftop and one for background. This final step ensured that the model could output dense segmentation masks at the pixel level delivering rooftop predictions across the entire image tile.

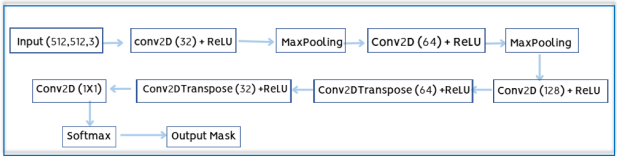

Figure 6. CNN architecture for image segmentation

4.2.3 Training Setup

The training of this architecture was conducted in a GPU-accelerated Colab environment. The Adam optimizer (Kingma and Ba, 2015) was selected for its ability to adapt learning rates dynamically, a feature particularly beneficial when training deep networks on heterogeneous image data. The objective function was binary cross-entropy, mathematically expressed as (Equation 1):

$$L = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] \qquad (1)$$

Where y_i represents the true class label of pixel i (0 for background, 1 for rooftop) and $\hat{y}_i$ is the predicted probability of belonging to the rooftop class. This formulation penalises incorrect predictions proportionally to their confidence, thereby encouraging the model to output precise probability maps.(Goodfellow et al., 2016)

Training was performed with a batch size of four image tiles balancing GPU memory limitations with training stability. Two setups were employed: a shorter run of 10 epoch which provided a baseline for model evaluation and a longer run of 50 epochs which allowed the network to refine its segmentation masks and improve boundary precision. Each epoch required approximately seven minutes resulting in several hours to sometime roughly half day of computation for the extended configuration, even with high GPU mode enabled. Over the course of training, the model progressively enhanced its ability to detect rooftops with early layers recognizing simple shapes and textures while deeper layers captured complex spatial patterns. The decoder in turn sharpened these representations into coherent segmentation masks

In practice, the encoder–decoder structure proved highly effective allowing the model not only to recognize rooftop regions across diverse urban scenes but also to reproduce them as detailed pixel-level masks. The careful design of the processing workflow combined with this CNN architecture and rigorous training setup provided a robust methodological foundation for extracting rooftops from high-resolution aerial imagery with both accuracy and generalizability.

5. Experiments and Result

The goal of the experiment was to determine whether a Convolutional Neural Network (CNN) can successfully extract building rooftops from high-resolution aerial images. The concept was to check if such a model could identify the patterns that define rooftops and distinguish them from the rest of the urban areas. As rooftop extraction is fundamentally a pixel-wise classification, the task was framed into a semantic segmentation exercise, with each pixel labelled as a rooftop or non-roof top.

The dataset used for this study was a collection of aerial images over Athens, Greece. To make the model focused on relevant structures, only images that contained rooftops were kept, while empty or non-building tiles were filtered out. The dataset provided only RGB bands, so the experiment was carried out using only the visible spectrum of light without multispectral cues.

A CNN-based encoder-decoder model was trained for 50 epochs. Checking the outcomes at various stages of training allowed monitoring of the network's progress in rooftop detection. After 10 epochs, the model's predictions were basic as some roofs were not detected, and those that were had no clear and smooth edges. However, the results got considerably better with further training. At epoch 50, the texture of roofs was more visible, segmentation masks were more complete, and overall predictions were closer to actual rooftop outlines.(Figure 7)

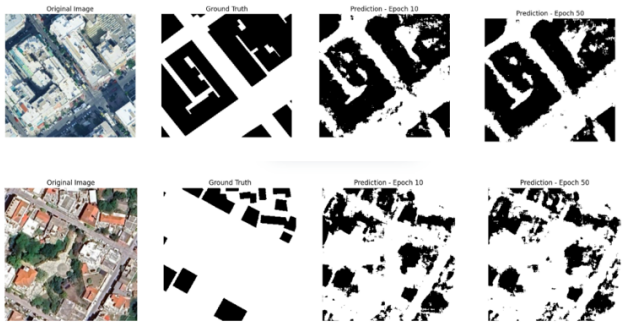

Figure 7. Good Detection

Despite this progress, the results were not perfect. Certain cases showed persistent error. In particular, rooftops were occasionally confused with roads due to similar colour tone and texture. Shadows, especially from tall structures and tree cover confused the model. These drawbacks implied that even though

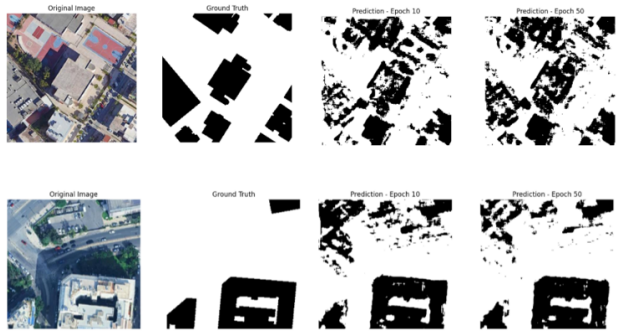

Figure 8. Bad Detection

RGB images might helpful in extracting roofs to a certain extent, they are insufficient to tackle the complexities of clustered urban areas. To a large extent, the experiment was a demonstration of the capabilities of CNNs to grasp both geometric and visual features of rooftops.(Figure 8)

However, the success rate is still limited by the quality and the range of the input data. The main finding is that longer training can improve accuracy but real progress will be dependent on better input sources as well as more advanced architectures.

6. Discussion

The results show the clear indications of both the potential and limitations of CNN-based rooftop extraction. The network showed that it can gradually learn rooftop patterns after the extended training. This advancement reflects the suitability of CNNs for the application of large-scale urban analysis, especially in cases of rooftops clearly visible and free of any obstruction. On the other hand, the difficulties encountered in misclassification and occluded areas underline the limitations of relying exclusively on RGB images. In dense cityscapes, where roads, parking lots, and rooftops often share similar visual features, it is unreasonable to expect colour and texture alone to resolve all ambiguities.

To address these issues, richer data sources and stronger models are required. Using multispectral or infrared bands would help the network to distinguish rooftops from surfaces with similar visual properties. In the same way adding elevation data such as digital surface model or digital terrain model could help the system to differentiate the structural information to separate flat ground from elevated rooftops. In a modelling perspective, advanced architectures such as U-Net++ and DeepLabV3+ are perfect choice for this type of task, as they enhance the capture of multiscale context and provide sharper object boundaries. The use of Vision Transformers also great because of their global attention mechanisms help in cases of occlusions and more complex background patterns. Moreover, hybrid modules like Multi-Scale Global Perceptron (MSGP) and Spatial Context Refinement Decoder (SCRD) can take the detection to a higher level by capturing rooftops of varying size and refining their boundaries.

7. Outlook

Future research should focus on enriching the input dataset and include a wider range of deep learning architectures. Use of multispectral or elevation data will contribute a significant solution, while data augmentation like rotation, flipping, and brightness adjustment will enable the model to generalise effectively. Benchmarking different segmentation model on the same dataset could compare their strength and weaknesses. The combination of advanced architecture with enriched data input can make rooftop extraction more reliable, scalable, and practical applications such as urban planning, disaster response and renewable energy assessment

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